
THE CORRELATION LIVES IN THE JOINT REGISTRATION

Entanglement on the Seat: No Signal as the Wall, the Correlation in Neither Local Record, and Bell's Bound as a Count of Local Strategies

The Closure Template of the Riemann Hypothesis Read on Two Parties; Fifty-Five Laws Proved in Core Lean 4 and Six Checks Executed in Fortran; a Witness of the Hardware Closure Series

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The correlation is the joint registration

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ABSTRACT. Entanglement and nonlocality are read here from the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. Two parties register outcomes, and their joint record counts the four outcome pairs. When one party relabels her outcome, her local bias and the correlation reverse while the other party's record does not move. A carrier that sends her bias to the offset from the line is equivariant, so her relabelling is carried onto the seat's reflection and the unbiased local record onto the line. We prove in core Lean 4, fifty-five laws with no axiom declared and no use of choice, that the far record is blind to every local relabelling, so no function of it returns a local choice: no signal passes, by the template's wall; that two joint records with the same unbiased local records on both sides carry opposite correlations, so the correlation is in neither local record and lives only in the joint registration; that every deterministic local strategy gives a CHSH value of exactly ± 2 , all sixteen counted, and every mixture of them obeys Bell's bound $|S| \leq 2$; and that a box with unbiased local records reaches 4 while signalling nothing. A Fortran twin runs six checks with zero failures: 100,000 local mixtures stay within 2; the singlet reaches $2\sqrt{2} = 2.828427$ at the optimal settings and no random setting exceeds it; its local records are unbiased to 10^{-16} whatever the far setting; and four experiments, from 1982 to 2023, measured CHSH values above 2 by 2.1 to 46.5 standard deviations and below $2\sqrt{2}$.

KEYWORDS *entanglement · nonlocality · Bell inequality · CHSH · no-signalling · Tsirelson bound · PR box · local hidden variables · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the registration read on two parties); every open row (the barrier, in two joint records with one pair of local records) · channel: kinetic · Lean: `SPHYS_Entanglement.lean` · Fortran: `Entanglement_Twin.f90`

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. Two distant parties register outcomes whose correlation no local account reproduces, and neither can use it to send the other a message. The seat reads both facts as one structure: a local relabelling is a fold the far record cannot see.

It claims four things.

- A local relabelling is the fold on the local record, and the unbiased local record is the line.
- No signal passes. The far record is even under every local relabelling and a local choice is odd, so the template's wall stands between them.
- The correlation is in neither local record. Two joint records with the same local records carry opposite correlations, so the correlation lives only in the joint registration.
- Bell's bound is a count of local strategies: each of the sixteen gives exactly ± 2 , so every mixture stays within 2.

What it refuses. A local record that carries the correlation. The engine proves there is none.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes

the weaker grade.

- No derivation of the Riemann Hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unacted.
- Interpretive schools from the literature enter only as placements the theorems sort, never as warrants; the only witness besides the mathematics is measured data.

Five terms.

- A *joint record* counts the four outcome pairs of two parties.
- A *local record* is one party's own outcome counts.
- The *correlation* is the agreements minus the disagreements.
- A *local strategy* fixes each party's outcome for each of its settings.
- A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on a local relabelling in Section 3, no signal in Section 4, the correlation in Section 5, Bell's bound in Section 6, and the quantum and measured values in Section 7.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; Light; Electromagnetism; Immanent Gravity; Spin and Statistics; Weak; Resonances; The Ferromagnet; Dark Matter; The Correlation Lives in the Joint Registration	the registration read on two parties: the unbiased local record the line, no signal as the wall, the correlation in the joint registration alone	WITNESSED · CROSSED · CONFIRMED on the image	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
every open row · the barrier	two worlds over one record	the series' witnesses; The Correlation Lives in the Joint Registration	two joint records with one pair of unbiased local records and opposite correlations	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (*value_on_image*).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (*equivariant_id*, *equivariant_comp*), and the fixed set is a functor on it (*fix_functorial*). An equivariant map into the seat, whose fixed set is the line (*fold_global_seat*), lands every

physical fixed point on the line (`equivariant_carrier_lands`). Here the physical involution is one party's relabelling of her outcome, which exchanges the rows of the joint record (`flipA_involutive`). The local carrier sends a joint record to $(1 + b_A, N)$, her bias b_A the offset from the line and the total count the height. It is equivariant (`local_carrier_equivariant`), and its landing on the fixed set is forced (`equivariant_carrier_lands`).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
a local relabelling	the fold	<code>local_carrier_equivariant, flipA_involutive</code>	A
an unbiased local record	the line	<code>unbiased_is_the_line</code>	A, D
the far record	even under the local fold	<code>bob_sees_no_flip</code>	A, D
no signal	the wall	<code>no_signal_from_a_local_choice</code>	D
the correlation	odd, in neither local record	<code>corr_odd, correlation_is_not_in_the_local_records</code>	A
a local strategy	± 2 exactly	<code>chsh_values, local_strategies_give_two</code>	B
a local model	Bell's bound	<code>bell_bound</code>	B
a maximal no-signalling box	4, signalling nothing	<code>pr_box_reaches_four, pr_box_signals_nothing</code>	E
the singlet	$2\sqrt{2}$	<code>computed</code>	C
the measured violations	above 2, below $2\sqrt{2}$	<code>corroboration</code>	F

3. A LOCAL RELABELLING IS THE FOLD

When Alice relabels her outcome, the joint record's rows exchange, which is an involution (`flipA_involutive`). Her bias reverses (`aliceBias_odd`) and the correlation reverses (`corr_odd`). The local carrier sends her relabelling to the seat's reflection (`local_carrier_equivariant`). Her local record lands on the line exactly when it is unbiased (`unbiased_is_the_line`). A maximally entangled pair puts every local record on the line: the twin finds the singlet's local records unbiased to 10^{-16} at 100,000 settings.

4. NO SIGNAL IS THE WALL

Bob's record, his own outcome counts, does not move when Alice relabels (`bob_sees_no_flip`). Whether Alice relabels is odd and Bob's record even, so no function of Bob's record returns her choice (`no_signal_from_a_local_choice`). This is no-signalling in exact form. Whatever Alice does on her side is a fold, and the far record is blind to every fold, as an even reading is blind to an odd target. The twin confirms it on the singlet: Bob's marginal is one half at every setting of Alice's.

5. THE CORRELATION LIVES IN THE JOINT REGISTRATION

Two joint records, one perfectly correlated and one perfectly anticorrelated, have the same unbiased record for Alice and the same for Bob, and opposite correlations. No function of the two local records returns the correlation (`correlation_is_not_in_the_local_records`). The correlation is a property of the joint registration alone, two worlds over one pair of local records. That is what entanglement is on the seat. Each local outcome is a keyed bit supplied by its own act, as the measurement paper proves. The correlation is the relation between the two acts, and it appears only when the two records are registered together.

6. BELL'S BOUND IS A COUNT OF LOCAL STRATEGIES

A local strategy fixes each party's outcome for each of two settings. The CHSH combination of its four correlations is ± 2 for every one of the sixteen strategies (`chsh_values`, `local_strategies_give_two`). Every mixture with nonnegative weights stays within twice its total weight (`bell_bound`). Bell's inequality (Bell, 1964; Clauser, Horne, Shimony and Holt, 1969) is this count and nothing more: an account that gives each party a local record of answers in advance cannot exceed 2. The twin draws 100,000 random mixtures, and every one stays within 2.

No-signalling alone allows more. A box whose outcomes agree unless both settings are 1 has unbiased local records for every setting and signals nothing (`pr_box_signals_nothing`), and it reaches 4 (`pr_box_reaches_four`; Popescu and Rohrlich, 1994).

7. THE QUANTUM VALUE AND THE MEASURED VALUES

The singlet's correlation at settings a and b is $-\cos(a - b)$. The twin reaches $2\sqrt{2} = 2.828427$ at the optimal settings, and no one of 100,000 random settings exceeds it, the bound Tsirelson proved (Tsirelson, 1980). The measured values:

Experiment	CHSH value	Above 2
Aspect, Grangier and Roger, 1982	2.697 ± 0.015	46.5σ
Weihs et al., 1998	2.73 ± 0.02	36.5σ
Hensen et al., 2015, loophole-free	2.42 ± 0.20	2.1σ
Storz et al., 2023, loophole-free	2.0747 ± 0.0033	22.6σ

Every one lies above the local count and below $2\sqrt{2}$.

The world's joint records are not mixtures of local ones. Its local records never signal.

8. WHAT THE PAIR CARRIES, AND WHAT IT DOES NOT

What it carries.

- The local relabelling as the fold and the unbiased local record as the line.
- No signal as the wall.
- The correlation in the joint registration alone.
- Bell's bound as a count of local strategies, and the measured violations.

What it does not carry.

- **The value $2\sqrt{2}$ from the template:** it is computed from the singlet and imported as Tsirelson's theorem.
- **A local record of the correlation:** there is none.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

9. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Signal:** a local operation on one party that changes the far party's own outcome counts. It breaks `bob_sees_no_flip` and `no_signal_from_a_local_choice` read physically.
- **F-Local:** a CHSH value above 2 reproduced by a model assigning each party its outcomes in advance. It breaks `bell_bound`.
- **F-Tsirelson:** a measured CHSH value above $2\sqrt{2}$ on quantum systems. It breaks the imported bound and the twin's computation of the singlet.

Nothing is owed on the row's channel.

Ledger status. Row 2 gains the registration read on two parties, WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains two joint records with one pair of local records.

METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author’s verification program, whose vocabulary is confined to this section. Its tokens:

- $[\perp]$, a field sealed;
- $[\perp T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[\cdot]$, the dot, which is no verdict.

Grades join at the weakest link. The engine’s statements are theorem grade about the chart. The mapping of physical states onto them is structural grade. Every measured figure is corroboration grade. Tsirelson’s bound is imported at its cited grade. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbd1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Entanglement.lean, 493 lines, SHA-256 23f2e80448c2b2b7..., fifty-five laws, no import and no axiom declared. Of the fifty-five, 22 depend on no axiom, 5 on propositional extensionality alone, 28 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system’s screen. Each of the fifty-five laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper’s seat, carriers and substrate verbatim.
- **Twin.** Entanglement_Twin.f90, 134 lines, SHA-256 480a2006e769de5d..., six checks and zero failures under -std=f2018 -O2 -fno-fast-math -ffp-contract=off. Measured inputs enter as data with their sources named in its header.

Why an integer engine binds continuous physics. Joint records are counts, integers by definition, and correlations are integer combinations of them, so the engine’s statements are exact; Bell’s bound is a statement about integer-weighted mixtures and so, by scaling, about every rational and every real mixture. The twin checks the quantum correlations directly.

The seventh. In the author’s register the six witnesses are six days and the ledger’s closure is the seventh, the Rest (Istawa). The pair is read at the Rest as two acts whose relation is written only when both are read together. In the Lean codex’s tokens its row renders $[\perp \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments’, cited by name. The theorems are the engine’s, printed with their dependencies in Appendix A. The twin’s figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author’s seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- PhysOS^T 1.0.5p in the public register (github.com/1000sapients/Trisduction, protocols/PhysOS/).

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M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Measurement Is the Registration, edition 1.0.0 (this series). M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_ENTANGLEMENT.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Entanglement.lean · the correlation lives in the joint
registration, read at the Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart:  $h = 2 \operatorname{Re} s$ ,  $t = \operatorname{Im} s$ ; the fold  $s \mapsto 1 - \bar{s}$  is  $(h, t) \mapsto (2 - h, t)$ , the critical line  $h = 1$ .

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier (resident).
Part III  the substrate (resident).
Part IV   Two local records and a local fold.
Part V    Bell's bound is a count of local strategies.
Part VI   The capstone.
-/
set_option autoImplicit false
namespace SPHYS.Entanglement

/-- ## Part I. The seat, the category of involutions, and the functor
of the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_land (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing
of Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
(fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
(reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ :
X → X} {σ : Y → Y}
{p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
Equivariant (g ∘ f) τ p := by
intro x
show g (f (τ x)) = p (g (f x))
rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points
to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y → Y}
(hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
by
rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
involutive : ∀ p, σ (σ p) = p
fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := {fold_involutive,
seat_fixed_line}

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
ι : W → Pt
lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involutions onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_land (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
∃ C : Carrier {x : X // τ x = x}, ∀ w, C.ι w = f w.1 :=
({fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
(fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=
fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
Value Z :=
fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_, rfl)
obtain (h, t) := p
have e' : 2 - h = 1 := e
show h = 1
omega

/-- ## Part II. The energy carrier: an energy, its rate, and the fold
-/

/-- An energy  $a + i \beta/2$ , as frequency and doubled rate; its mode's norm
moves as  $e^{\beta t}$ . -/
structure Energy where
freq : Int
rate : Int
deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm  $e^{\beta t}$  is unchanged at every integer time t,
i.e.  $\beta t = 0$  for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

```

```

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 -  $\beta$ , t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d)  $\leftrightarrow$  a = c  $\wedge$  b = d :=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
    fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [psi_phi E, psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E  $\leftrightarrow$  IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)  $\leftrightarrow$  b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E)  $\leftrightarrow$  IsReal E :=
  by
  obtain (a, b) := E
  show 1 - b = 1  $\leftrightarrow$  b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E  $\leftrightarrow$  IsReal E :=
  by
  constructor
  · intro h; have h1 := h.1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h, Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p  $\leftrightarrow$  Stationary (psi p) :=
  by
  rw [stationary_iff_real, phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp :  $\neg$  OnLine p) (a : Int) :
  phi (a, 0)  $\neq$  p :=
  fun e => hp (e  $\rightarrow$  measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
   $\exists$  C : Carrier { E : Energy // E.mirror = E },  $\forall$  w, C.1 w = phi w.1 :=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}

/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0  $\wedge$  c.colour = 0  $\wedge$  c.b3 = 0
 $\wedge$  c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c  $\leftrightarrow$  Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)  $\leftrightarrow$  (q = 0  $\wedge$  k = 0  $\wedge$  b = 0  $\wedge$  l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3  $\wedge$  (bar c).colour = -c.colour  $\wedge$  (bar c).b3 =
-c.b3  $\wedge$ 
  (bar c).lepton = -c.lepton)  $\wedge$  (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit { $\beta$  : Type} (g : Energy  $\rightarrow$   $\beta$ ) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c  $\wedge$  tmirror c = c)  $\leftrightarrow$  (Neutral c  $\wedge$  IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The energy carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
  by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
  fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by

```

```

show 1 + c.q3 = 1
omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. Two local records and a local fold -/

/-- A joint record of two parties: counts of the four outcome pairs
(++, ++, --, --). -/
structure Table where
  pp : Int
  pm : Int
  mp : Int
  mm : Int
  deriving DecidableEq, Repr

/-- Alice relabels her outcome: n(a, b) ↦ n(-a, b). -/
def flipA (t : Table) : Table := (t.mp, t.mm, t.pp, t.pm)

theorem flipA_involutive (t : Table) : flipA (flipA t) = t := rfl

def corr (t : Table) : Int := t.pp - t.pm - t.mp + t.mm
def aliceBias (t : Table) : Int := (t.pp + t.pm) - (t.mp + t.mm)
def bobRecord (t : Table) : Int × Int := (t.pp + t.mp, t.pm + t.mm)

theorem corr_odd (t : Table) : corr (flipA t) = - corr t := by
  show t.mp - t.mm - t.pp + t.pm = -(t.pp - t.pm - t.mp + t.mm); omega

theorem aliceBias_odd (t : Table) : aliceBias (flipA t) = - aliceBias t
:= by
  show (t.mp + t.mm) - (t.pp + t.pm) = -(t.pp + t.mp) - (t.mp +
  t.mm); omega

/-! NO SIGNAL: Alice's local operation leaves Bob's record unchanged.
-/
theorem bob_sees_no_flip (t : Table) : bobRecord (flipA t) = bobRecord
t := by
  show ((t.mp + t.pp, t.mm + t.pm) : Int × Int) = (t.pp + t.mp, t.pm +
  t.mm)
  rw [Int.add_comm t.mp t.pp, Int.add_comm t.mm t.pm]

/-- The local carrier: Alice's bias is the offset from the line, the
total the height. -/
def localSeat (t : Table) : Pt := (1 + aliceBias t, t.pp + t.pm + t.mp
+ t.mm)

theorem local_carrier_equivariant : Equivariant localSeat flipA fold :=
by
  intro t
  show ((1 + aliceBias (flipA t), t.mp + t.mm + t.pp + t.pm) : Pt) = (2
- (1 + aliceBias t), t.pp + t.pm + t.mp + t.mm)
  rw [aliceBias_odd, pe]; exact (by omega, by omega)

/-- An unbiased local record is the line. -/
theorem unbiased_is_the_line (t : Table) : OnLine (localSeat t) ↔
aliceBias t = 0 := by
  show 1 + aliceBias t = 1 ↔ aliceBias t = 0
  exact (fun h => by omega, fun h => by omega)

/-! THE CORRELATION IS NOT IN THE LOCAL RECORDS: two joint records with
the same unbiased local
records on both sides carry opposite correlations, so no function
of the local records returns
the correlation. -/
theorem correlation_is_not_in_the_local_records (g : Int → Int × Int
→ Int) :
  ∀ t : Table, g (aliceBias t) (bobRecord t) = corr t := by
  intro hg
  have h1 := hg (1, 0, 0, 1)
  have h2 := hg (0, 1, 1, 0)
  change g 0 (1, 1) = 2 at h1; change g 0 (1, 1) = -2 at h2
  omega

/-! NO SIGNAL, AS A WALL: whether Alice flipped is odd, Bob's record is
even, and no function of
Bob's record returns Alice's choice. -/
theorem no_signal_from_a_local_choice (g : Int × Int → Bool) (t :
Table) :
  ∀ b : Bool, g (bobRecord (if b then flipA t else t)) = b := by
  intro hg
  have h1 : g (bobRecord (flipA t)) = true := hg true
  have h2 : g (bobRecord t) = false := hg false
  rw [bob_sees_no_flip] at h1
  rw [h1] at h2; exact Bool.noConfusion h2

/-! ## Part V. Bell's bound is a count of local strategies -/

def sgn (b : Bool) : Int := if b then 1 else -1

/-- The CHSH value of one deterministic local strategy: outcomes fixed
for each setting. -/
def chsh (a0 a1 b0 b1 : Bool) : Int :=
  sgn a0 * sgn b0 + sgn a0 * sgn b1 + sgn a1 * sgn b0 - sgn a1 * sgn b1

/-- EVERY LOCAL STRATEGY GIVES ±2: all sixteen counted. -/
theorem local_strategies_give_two :
  ([false, true].all fun a0 => [false, true].all fun a1 => [false,
  true].all fun b0 =>
  [false, true].all fun b1 => chsh a0 a1 b0 b1 == 2 || chsh a0 a1
  b0 b1 == -2) = true := by
  decide

theorem chsh_values (a0 a1 b0 b1 : Bool) : chsh a0 a1 b0 b1 = 2 ∨ chsh
a0 a1 b0 b1 = -2 := by
  cases a0 <|> cases a1 <|> cases b0 <|> cases b1 <|> decide

/-! A local model is a mixture of strategies with nonnegative weights.
-/
def mixS : List (Int × (Bool × Bool × Bool × Bool)) → Int
| [] => 0
| (w, (a0, a1, b0, b1)) :: rest => w * chsh a0 a1 b0 b1 + mixS rest

def mixW : List (Int × (Bool × Bool × Bool × Bool)) → Int
| [] => 0
| (w, _) :: rest => w + mixW rest

/-! BELL'S BOUND: every local mixture obeys |S| ≤ 2 times its total
weight. -/
theorem bell_bound (l : List (Int × (Bool × Bool × Bool × Bool))) (hw :
∀ e ∈ l, 0 ≤ e.1) :
  -2 * mixW l ≤ mixS l ∧ mixS l ≤ 2 * mixW l := by
  induction l with
  | nil => exact (by decide, by decide)
  | cons e rest ih =>
    obtain (w, a0, a1, b0, b1) := e
    have h0 : 0 ≤ w := hw _ (List.Mem.head _)
    have ih' := ih (fun x hx => hw x (List.Mem.tail _ hx))
    show -2 * (w + mixW rest) ≤ w * chsh a0 a1 b0 b1 + mixS rest ∧ w *
    chsh a0 a1 b0 b1 + mixS rest ≤ 2 * (w + mixW rest)
    rcases chsh_values a0 a1 b0 b1 with h | h <|> rw [h] <|>
    constructor <|> omega

/-! THE PR BOX: the outcomes agree unless both settings are 1, where
they disagree; scaled by two,
each allowed pair carries weight 1, so each correlation reads ±2
and the scaled CHSH sum 8 is a
CHSH value of 4. Its local records are unbiased for every setting.
-/
def prP (x y a b : Bool) : Int := if (a != b) = (x && y) then 1 else 0
def prE (x y : Bool) : Int := prP x y true true - prP x y true false -
prP x y false true + prP x y false false

theorem pr_box_reaches_four : prE false false + prE false true + prE
true false - prE true true = 2 * 4 := by decide

theorem pr_box_signals_nothing :
  ([false, true].all fun x => [false, true].all fun y => [false,
  true].all fun a =>
  (prP x y a true + prP x y a false == prP x (!y) a true + prP x
  (!y) a false) &&
  (prP y x true a + prP y x false a == prP (!y) x true a + prP (!y)
  x false a)) = true := by
  decide

/-! ## Part VI. The capstone -/

/-! THE CORRELATION LIVES IN THE JOINT REGISTRATION. A local
relabeling is the fold on the local
record and the unbiased record the line; the far record is blind to
it, so no signal passes; the
correlation is odd and no function of the local records returns it;
every local strategy gives
±2 and every local mixture obeys Bell's bound; and a no-signalling
box reaches 4. -/
theorem correlation_lives_in_the_joint_registration :
  Equivariant localSeat flipA fold ∧
  (∀ t : Table, OnLine (localSeat t) ↔ aliceBias t = 0) ∧
  (∀ t : Table, bobRecord (flipA t) = bobRecord t ∧ corr (flipA t) = -
  corr t) ∧
  (∀ g : Int → Int × Int → Int, ∀ t : Table, g (aliceBias t)
  (bobRecord t) = corr t) ∧
  (∀ (g : Int × Int → Bool) (t : Table), ∀ b : Bool, g (bobRecord
  (if b then flipA t else t)) = b) ∧
  (∀ l : List (Int × (Bool × Bool × Bool × Bool)), (∀ e ∈ l, 0 ≤ e.1)
  →
  -2 * mixW l ≤ mixS l ∧ mixS l ≤ 2 * mixW l) ∧
  prE false false + prE false true + prE true false - prE true true =
  2 * 4 :=
  (local_carrier_equivariant, unbiased_is_the_line, fun t =>
  (bob_sees_no_flip t, corr_odd t),
  correlation_is_not_in_the_local_records,
  no_signal_from_a_local_choice, bell_bound, pr_box_reaches_four)

```

```

end SPHYS.Entanglement

#print axioms SPHYS.Entanglement.pe
#print axioms SPHYS.Entanglement.fold_involutive
#print axioms SPHYS.Entanglement.seat_fixed_line
#print axioms SPHYS.Entanglement.reg_lands
#print axioms SPHYS.Entanglement.reg_fixes_iff
#print axioms SPHYS.Entanglement.reg_forgets_side
#print axioms SPHYS.Entanglement.least_erasure_iff_value
#print axioms SPHYS.Entanglement.equivariant_id
#print axioms SPHYS.Entanglement.equivariant_comp
#print axioms SPHYS.Entanglement.fix_functorial
#print axioms SPHYS.Entanglement.fold_global_seat
#print axioms SPHYS.Entanglement.equivariant_carrier_lands
#print axioms SPHYS.Entanglement.value_on_image
#print axioms SPHYS.Entanglement.kinetic_crossing
#print axioms SPHYS.Entanglement.off_locus_pair
#print axioms SPHYS.Entanglement.ee
#print axioms SPHYS.Entanglement.mirror_involutive
#print axioms SPHYS.Entanglement.psi_phi
#print axioms SPHYS.Entanglement.phi_psi
#print axioms SPHYS.Entanglement.phi_injective
#print axioms SPHYS.Entanglement.phi_equivariant
#print axioms SPHYS.Entanglement.mirror_fixed_iff_real
#print axioms SPHYS.Entanglement.phi_line_iff_real
#print axioms SPHYS.Entanglement.stationary_iff_real
#print axioms SPHYS.Entanglement.line_is_stationary
#print axioms SPHYS.Entanglement.measured_on_line
#print axioms SPHYS.Entanglement.no_measurement_off_line

'SPHYS.Entanglement.pe' does not depend on any axioms
'SPHYS.Entanglement.fold_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.seat_fixed_line' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.reg_lands' does not depend on any axioms
'SPHYS.Entanglement.reg_fixes_iff' depends on axioms: [propxet]
'SPHYS.Entanglement.reg_forgets_side' does not depend on any axioms
'SPHYS.Entanglement.least_erasure_iff_value' depends on axioms:
[propxet]
'SPHYS.Entanglement.equivariant_id' does not depend on any axioms
'SPHYS.Entanglement.equivariant_comp' does not depend on any axioms
'SPHYS.Entanglement.fix_functorial' does not depend on any axioms
'SPHYS.Entanglement.fold_global_seat' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.equivariant_carrier_lands' depends on axioms:
[propxet, Quot.sound]
'SPHYS.Entanglement.value_on_image' does not depend on any axioms
'SPHYS.Entanglement.kinetic_crossing' does not depend on any axioms
'SPHYS.Entanglement.off_locus_pair' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.ee' does not depend on any axioms
'SPHYS.Entanglement.mirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.psi_phi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Entanglement.phi_psi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Entanglement.phi_injective' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.phi_equivariant' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.mirror_fixed_iff_real' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.phi_line_iff_real' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.stationary_iff_real' depends on axioms: [propxet]
'SPHYS.Entanglement.line_is_stationary' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.measured_on_line' does not depend on any axioms
'SPHYS.Entanglement.no_measurement_off_line' does not depend on any
axioms
'SPHYS.Entanglement.energy_carrier_lands' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.bar_involutive' does not depend on any axioms

#print axioms SPHYS.Entanglement.energy_carrier_lands
#print axioms SPHYS.Entanglement.bar_involutive
#print axioms SPHYS.Entanglement.bar_fixed_iff_neutral
#print axioms SPHYS.Entanglement.odd_charges_even_energy
#print axioms SPHYS.Entanglement.gravity_reads_no_charge_bit
#print axioms SPHYS.Entanglement.involutions_commute
#print axioms SPHYS.Entanglement.tmirror_involutive
#print axioms SPHYS.Entanglement.joint_fixed_iff
#print axioms SPHYS.Entanglement.charge_carrier_equivariant
#print axioms SPHYS.Entanglement.pair_lands_once
#print axioms SPHYS.Entanglement.tmirror_equivariant
#print axioms SPHYS.Entanglement.neutral_charge_on_line
#print axioms SPHYS.Entanglement.pair_on_the_edges
#print axioms SPHYS.Entanglement.stable_lands
#print axioms SPHYS.Entanglement.flipA_involutive
#print axioms SPHYS.Entanglement.corr_odd
#print axioms SPHYS.Entanglement.aliceBias_odd
#print axioms SPHYS.Entanglement.bob_sees_no_flip
#print axioms SPHYS.Entanglement.local_carrier_equivariant
#print axioms SPHYS.Entanglement.unbiased_is_the_line
#print axioms
SPHYS.Entanglement.correlation_is_not_in_the_local_records
#print axioms SPHYS.Entanglement.no_signal_from_a_local_choice
#print axioms SPHYS.Entanglement.local_strategies_give_two
#print axioms SPHYS.Entanglement.chsh_values
#print axioms SPHYS.Entanglement.bell_bound
#print axioms SPHYS.Entanglement.pr_box_reaches_four
#print axioms SPHYS.Entanglement.pr_box_signals_nothing
#print axioms
SPHYS.Entanglement.correlation_lives_in_the_joint_registration

'SPHYS.Entanglement.bar_fixed_iff_neutral' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.odd_charges_even_energy' does not depend on any
axioms
'SPHYS.Entanglement.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.Entanglement.involutions_commute' does not depend on any axioms
'SPHYS.Entanglement.tmirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.joint_fixed_iff' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.charge_carrier_equivariant' depends on axioms:
[propxet, Quot.sound]
'SPHYS.Entanglement.pair_lands_once' does not depend on any axioms
'SPHYS.Entanglement.tmirror_equivariant' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.neutral_charge_on_line' depends on axioms:
[propxet, Quot.sound]
'SPHYS.Entanglement.pair_on_the_edges' does not depend on any axioms
'SPHYS.Entanglement.stable_lands' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.flipA_involutive' does not depend on any axioms
'SPHYS.Entanglement.corr_odd' depends on axioms: [propxet, Quot.sound]
'SPHYS.Entanglement.aliceBias_odd' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.bob_sees_no_flip' depends on axioms: [propxet]
'SPHYS.Entanglement.local_carrier_equivariant' depends on axioms:
[propxet, Quot.sound]
'SPHYS.Entanglement.unbiased_is_the_line' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.correlation_is_not_in_the_local_records' depends on
axioms: [propxet, Quot.sound]
'SPHYS.Entanglement.no_signal_from_a_local_choice' depends on axioms:
[propxet]
'SPHYS.Entanglement.local_strategies_give_two' does not depend on any
axioms
'SPHYS.Entanglement.chsh_values' does not depend on any axioms
'SPHYS.Entanglement.bell_bound' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Entanglement.pr_box_reaches_four' does not depend on any axioms
'SPHYS.Entanglement.pr_box_signals_nothing' does not depend on any
axioms
'SPHYS.Entanglement.correlation_lives_in_the_joint_registration'
depends on axioms: [propxet, Quot.sound]

```

APPENDIX B: ENTANGLEMENT_TWIN.F90 AND ITS RUN

```

! Entanglement_Twin.f90 · the executed twin of SPHYS_Entanglement.lean
· Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Entanglement_Twin.f90
! Measured CHSH values: Aspect, Grangier and Roger, PRL 49 (1982) 91:
2.697 +- 0.015; Weihs et al., PRL 81
! (1998) 5039: 2.73 +- 0.02; Hensen et al., Nature 526 (2015) 682: 2.42
+- 0.20; Storz et al., Nature 617
! (2023) 265: 2.0747 +- 0.0033.
! A the local carrier and the wall of no signal; B every local strategy
gives +2, every mixture <= 2;
! C the singlet reaches 2 sqrt 2 and no more; D its local records are
unbiased at every setting;

```

```

! E the PR box reaches 4 and signals nothing; F the measured
violations.
program entanglement_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: PI = 3.141592653589793_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20261001_i8
call block_a(); call block_b(); call block_c(); call block_d(); call
block_e(); call block_f()

```

```

write(*,'(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
include 'twin_common.inc'
subroutine block_a()
integer :: pp, pm, mp, mm, n, ok
write(*,'(a)') 'A · Alice relabels her outcome: her bias and the
correlation flip, Bob's record does not move'
n = 0; ok = 0
do pp = 0, 6
do pm = 0, 6
do mp = 0, 6
do mm = 0, 6
n = n + 1
if ((mp - mm - pp + pm == -(pp - pm - mp + mm)) .and. (mp +
pp == pp + mp) .and. (mm + pm == pm + mm) .and. &
(1 + ((mp + mm) - (pp + pm)) == 2 - (1 + ((pp + pm) -
(mp + mm)))))) ok = ok + 1
end do
end do
end do
call check('2401 joint records: correlation odd, Bob's record
even, the local carrier equivariant', ok == n)
end subroutine block_a
subroutine block_b()
integer :: a0, a1, b0, b1, n2, k, j
real(dp) :: w(16), s(16), tot, ss
integer :: idx
logical :: okmix
write(*,'(a)') 'B · local strategies: sixteen deterministic, every
one at +-2; random mixtures within the bound'
n2 = 0; idx = 0
do a0 = -1, 1, 2
do a1 = -1, 1, 2
do b0 = -1, 1, 2
do b1 = -1, 1, 2
idx = idx + 1
s(idx) = real(a0*b0 + a0*b1 + a1*b0 - a1*b1, dp)
if (abs(s(idx)) == 2.0_dp) n2 = n2 + 1
end do
end do
end do
end do
okmix = .true.
do k = 1, 100000
tot = 0.0_dp; ss = 0.0_dp
do j = 1, 16
w(j) = rnd(); tot = tot + w(j); ss = ss + w(j)*s(j)
end do
if (abs(ss/tot) > 2.0_dp + 1.0e-12_dp) okmix = .false.
end do
call check('16 of 16 strategies give |S| = 2; 100000 mixtures obey
|S| <= 2', n2 == 16 .and. okmix)
end subroutine block_b
real(dp) function chsh_q(a0, a1, b0, b1) result(s)
real(dp), intent(in) :: a0, a1, b0, b1
s = -cos(a0 - b0) - cos(a0 - b1) - cos(a1 - b0) + cos(a1 - b1)
end function chsh_q
subroutine block_c()
integer :: k
real(dp) :: smax, sopt
write(*,'(a)') 'C · the singlet, E = -cos(a - b): CHSH at most 2
sqrt 2, reached at the optimal settings'
smax = 0.0_dp
do k = 1, 100000
smax = max(smax, abs(chsh_q(2.0_dp*PI*rnd(), 2.0_dp*PI*rnd(),
2.0_dp*PI*rnd(), 2.0_dp*PI*rnd())))
end do

```

A · Alice relabels her outcome: her bias and the correlation flip,
Bob's record does not move
PASS 2401 joint records: correlation odd, Bob's record even, the
local carrier equivariant
B · local strategies: sixteen deterministic, every one at +-2; random
mixtures within the bound
PASS 16 of 16 strategies give |S| = 2; 100000 mixtures obey |S| <= 2
C · the singlet, E = -cos(a - b): CHSH at most 2 sqrt 2, reached at the
optimal settings
optimum 2.828427125 ; largest of 100000 random settings 2.825993932
PASS the singlet reaches 2.828427 = 2 sqrt 2 and no random setting
exceeds it
D · the singlet's local records: P(a, b) = (1 - a b cos theta)/4; every
marginal 1/2

```

sopt = abs(chsh_q(0.0_dp, PI/2.0_dp, PI/4.0_dp, -PI/4.0_dp))
write(*,'(a,f12.9,a,f12.9)') ' optimum ', sopt, ' ; largest of
100000 random settings ', smax
call check('the singlet reaches 2.828427 = 2 sqrt 2 and no random
setting exceeds it', &
abs(sopt - 2.0_dp*sqrt(2.0_dp)) < 1.0e-14_dp .and. smax
<= 2.0_dp*sqrt(2.0_dp) + 1.0e-12_dp)
end subroutine block_c
subroutine block_d()
integer :: k
real(dp) :: th, pa, pb, worst
write(*,'(a)') 'D · the singlet's local records: P(a, b) = (1 - a
b cos theta)/4; every marginal 1/2'
worst = 0.0_dp
do k = 1, 100000
th = 2.0_dp*PI*rnd()
pa = (1.0_dp - cos(th))/4.0_dp + (1.0_dp + cos(th))/4.0_dp
pb = (1.0_dp - cos(th))/4.0_dp + (1.0_dp + cos(th))/4.0_dp
worst = max(worst, abs(pa - 0.5_dp), abs(pb - 0.5_dp))
end do
call check('100000 settings: both local records unbiased to 1e-16,
whatever the far setting', worst < 1.0e-15_dp)
end subroutine block_d
subroutine block_e()
integer :: x, y, a, b, s, ok
integer :: p(0:1, 0:1, 0:1, 0:1)
write(*,'(a)') 'E · the PR box: outcomes agree unless both settings
are 1'
do x = 0, 1
do y = 0, 1
do a = 0, 1
do b = 0, 1
p(x, y, a, b) = merge(1, 0, ieor(a, b) == iand(x, y))
end do
end do
end do
end do
s = 0; ok = 0
do x = 0, 1
do y = 0, 1
s = s + merge(-1, 1, x == 1 .and. y == 1)*(p(x, y, 1, 1) - p(x,
y, 1, 0) - p(x, y, 0, 1) + p(x, y, 0, 0))
do a = 0, 1
if (p(x, y, a, 0) + p(x, y, a, 1) == p(x, 1 - y, a, 0) + p(x,
1 - y, a, 1)) ok = ok + 1
if (p(y, x, 0, a) + p(y, x, 1, a) == p(1 - y, x, 0, a) + p(1 -
y, x, 1, a)) ok = ok + 1
end do
end do
end do
call check('the PR box reaches CHSH 4 (scaled sum 8) and every
marginal ignores the far setting', s == 8 .and. ok == 16)
end subroutine block_e
subroutine block_f()
real(dp) :: sv(4), se(4), r2
integer :: k, ok
write(*,'(a)') 'F · measured CHSH values, each above the local
bound 2 and below 2 sqrt 2'
sv = [2.697_dp, 2.73_dp, 2.42_dp, 2.0747_dp]; se = [0.015_dp,
0.02_dp, 0.20_dp, 0.0033_dp]
r2 = 2.0_dp*sqrt(2.0_dp); ok = 0
do k = 1, 4
write(*,'(a,f7.4,a,f7.4,a,f6.1,a)') ' S = ', sv(k), ' +- ',
se(k), ' : ', (sv(k) - 2.0_dp)/se(k), ' sigma above 2'
if (sv(k) > 2.0_dp .and. sv(k) < r2) ok = ok + 1
end do
call check('Aspect 1982, Weihs 1998, Hensen 2015, Storz 2023: all
above 2, all below 2.8284', ok == 4)
end subroutine block_f
end program entanglement_twin

```

PASS 100000 settings: both local records unbiased to 1e-16, whatever
the far setting
E · the PR box: outcomes agree unless both settings are 1
PASS the PR box reaches CHSH 4 (scaled sum 8) and every marginal
ignores the far setting
F · measured CHSH values, each above the local bound 2 and below 2 sqrt
2
S = 2.6970 +- 0.0150 : 46.5 sigma above 2
S = 2.7300 +- 0.0200 : 36.5 sigma above 2
S = 2.4200 +- 0.2000 : 2.1 sigma above 2
S = 2.0747 +- 0.0033 : 22.6 sigma above 2
PASS Aspect 1982, Weihs 1998, Hensen 2015, Storz 2023: all above 2,
all below 2.8284
BATTERY-JSON: {"checks":6,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

```
Phys0ST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caeee842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 Phys0S Proofs read from II.10, record
lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6
retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_T0E_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB
One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure
0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB
RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin
0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 55
JUDGMENT refused 55 of 55 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True
```